Unit 11, Lesson 1: Adding Rational Numbers

Benchmark

MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.

Learning Target

- [ ] I can fluently add rational numbers.

11.1.1 Warm-Up: Landscape Architect

Janice Nicol, a landscape architect, works to design natural systems and outdoor spaces, like trails and parks, for campuses and recreational facilities. Her job is both creative and technical. Creativity is expressed through drawing sketches both by hand and by using computer design programs and imaging. The technical side of her job involves knowing things such as land grading, water calculations, and natural systems.

Janice mentions that she regularly collaborates and works together with others while doing her job.

1. Why do you think it is important to be able to collaborate when doing a job?

2. How can you build your collaboration skills?

11.1.2 Guided Instruction: Adding Positive Rational Numbers

1. Adrian added 5.25 and 6.4. His work is shown.

Adrian's Work
$\begin{array}{r} 5.25 \\ +\ 6.4 \\ \hline 5.89 \end{array}$

a. Write an explanation to Adrian detailing any mistakes he made and how he can correct his work.

b. What is the sum of 5.25 and 6.4?

When finding the sum of decimal numbers, add values with the same place value.

2. Johannes added $\frac{1}{8}$ and $\frac{3}{4}$. Part of her work is shown.

a. Complete Johannes' work by filling in the boxes.

Johannes' Answer
$\frac{1}{8} + \frac{3}{4} =$
$\frac{1}{8} + \frac{3}{4}\left(\frac{2}{2}\right) =$
$\frac{1}{8} + \frac{\square}{\square} = \frac{\square}{\square}$

b. Complete the statements by choosing a word or number from the bank. Not all words and numbers may be used, and some may be used more than once.

Word Bank

2	4	8
numerators	common denominator	
one half	fourths	seven eighths

To find the sum of $\frac{1}{8}$ and $\frac{3}{4}$, Johannes first rewrote each fraction using a ______________________.

The ______________________ of 8 and 4 that Johannes used was _____. Then, she added the two __________________ together to find the total number of __________________.

11.1.3 Try It: Adding Positive Rational Numbers

1. Match pairs of expressions that have the same sum. Write the equivalent expressions on the lines provided.

$\frac{3}{4}+\frac{4}{8}$	$\frac{2}{3}+\frac{3}{5}$	$\frac{2}{3}+\frac{7}{18}$
$\frac{5}{9}+\frac{1}{2}$	$\frac{15}{15}+\frac{8}{30}$	$\frac{12}{16}+\frac{1}{2}$

_____________ = _____________

_____________ = _____________

_____________ = _____________

11.1.4 Refresher: Operations With Integers

1. Work with your partner to complete the table. The first addition expression has been done for you.

Expression	Absolute Value of First Addend	Absolute Value of Second Addend	Addend With Greater Absolute Value	Sign of Sum	Sum
$(^{-}10)+6$	10	6	$^{-}10$	–	$^{-}4$
$10+(^{-}6)$					
$(^{-}10)+(^{-}6)$					
$10+6$					

2. How did the size of the absolute value for each addend determine the sign of the sum?

3. Explain how finding the sum for the first two expressions was different from finding the sum for the last two expressions.

4. How were the sums for $10 + (^-6)$ and $(^-10) + 6$ similar?

5. How were the sums for $(^-10) + (^-6)$ and $10 + 6$ different?

6. Complete the statements using the words *same* or *different* to describe the rules for finding the sums of signed numbers.

When two addends have the _______________ sign, the sum will have the sign of the addends.

When two addends have _______________ signs, the sum will have the sign of the addend with the greater absolute value.

7. Apply the rules for integers to these expressions with rational numbers. Find the sum of each expression. Then, verify your answer using a calculator.

Addition Expression	Sum When Using the Rule	Sum With a Calculator	Does the Rule Work?
$(^-0.92) + 50.6$			
$4.95 + 3.05$			
$\left(-\frac{4}{5}\right) + \left(\frac{22}{25}\right)$			
$1.25 + (^-6.45)$			
$\left(-\frac{2}{3}\right) + \left(-\frac{9}{20}\right)$			
$(^-17.01) + (^-2.99)$			

8. Complete the sentence.

The sign rules for adding integers are
○ the same
○ not the same
as the rules for adding rational numbers.

11.1.5 Guided Practice: Applying Signed Number Rules to Rational Numbers

1. The blanks in the addition equation represent missing values. Fill in the blanks for the addition equation below to make a true statement.

$$-12\frac{3}{4} + \left(-5\frac{\boxed{}}{6}\right) = -18\frac{\boxed{}}{12}$$

2. Find the sum for each addition expression.

Addition Expression	Sum
$\frac{4}{5} + \left(-\frac{3}{7}\right)$	
$-9\frac{1}{2} + \left(-14\frac{1}{8}\right)$	

3. The blanks in the addition equation represent digits in the values.

   a. Fill in the blanks for the addition equation below to create a true equation.

$$-16.537 + 6.1\boxed{}8 = -10.\boxed{}59$$

   b. Describe the process you used to determine the values for each of the blanks above.

   c. Discuss your process with your partner. Write a summary of their process. Have your partner initial to show that your summary accurately expressed their process.

Your Partner's Process
Partner's Initials ______

11.1.6 Your Turn!

1. Find the sum for each expression.

a. $-1\frac{1}{2} + \left(-\frac{1}{8}\right)$

b. $-12.37 + 18.161$

c. $-\frac{2}{5} + (-7\frac{1}{4})$

d. $-4.48 + 9.366$

e. $-233.24 + 25.8$

f. $-4\frac{1}{7} + 17\frac{2}{9}$

11.1.7 Wrap-Up: Ticket Out the Door

1. Choose a value from Row A and a value from Row B that, when added, will result in a negative answer. Then, calculate the sum.

Row A

$-1\frac{3}{5}$	$-\frac{1}{8}$	$-2\frac{1}{4}$

Row B

$1\frac{1}{2}$	$2\frac{6}{25}$	$\frac{19}{40}$

Row A		Row B		Sum
	+		=	